



**Project Proposal Plan**  
**Rain Prediction Challenge**

**IPS Program Spring 2026**

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# **Project Objectives**

## **1. Data Exploration and Cleaning**

- Perform exploratory data analysis;
- Visualize variable distributions;
- Analyze correlations between variables;
- Detect missing values;
- Impute or remove missing data;
- Identify outliers or inconsistent values.

## **2. Feature Engineering**

- Transform numerical variables;
- Encode categorical variables;
- Create interaction features;
- Derive new meteorological indicators;
- Normalize or standardize variables when needed.

## **3. Predictive Modeling**

- Logistic regression
- Decision trees
- Random forests
- Gradient boosting :XGBoost, LightGBM or CAT-Boost
- Support vector machine
- Deep learning model : GRU

## **4. Model Validation and Hyperparameter Tuning**

- Train/Test split
- Cross-Validation
- Hyperparameter tuning using GridSearch, RandomSearch or Optuna
- Overfitting analysis
- Probability calibration if needed.

## **5. Model Evaluation Metrics**

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC
- Precision-recall curves, especially if the classes are imbalanced
- Confusion matrix

## **6. Model Interpretation and Error Analysis**

- Feature importance;
- SHAP values;
- Partial dependence plots;
- Error analysis on false positives and false negatives.

## **7. Application & Visualization**

- Enter weather conditions
- Obtain a predicted probability of rain

## Project Plan :

| Project Phases | Tasks                                                                                                                                                                                                                                                              |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase 1        | <ul style="list-style-type: none"><li>- EDA</li><li>- Data Cleaning &amp; Preprocessing</li><li>- Feature Engineering</li><li>- Random Forests</li><li>- SVM Model</li><li>- Models Evaluation Analysis &amp; Interpretation</li><li>- Follow Up Meeting</li></ul> |
| Phase 2        | <ul style="list-style-type: none"><li>- Gradient Boosting Models</li><li>- GRU Deep Learning Model</li><li>- Models Evaluation Analysis &amp; Interpretation</li><li>- Application &amp; Visualization</li><li>- Follow Up Meeting</li></ul>                       |
| Phase 3        | <ul style="list-style-type: none"><li>- Project Overall Review &amp; Analysis</li><li>- Report &amp; Documentation</li><li>- Presentation</li></ul>                                                                                                                |